

STARS AND STAR GROUPS

Bright stars

The stars seen at night have different brightnesses. Sirius, the brightest, appears about 1,000 times brighter than the faintest seen with the naked eye. The ancient Greek astronomer, Hipparchus, grouped the stars into “importance” categories, 1 being the brightest and 6 the faintest. Today we call these importances “apparent magnitudes.” Each unit of apparent magnitude is 2.51 times fainter than the one before. A star’s absolute magnitude is the apparent magnitude it would have if at a standard distance of 32.6 light-years.

BRIGHTEST STARS					
Name	Constellation	Apparent magnitude	Absolute magnitude	Distance from Earth (in light-years)	Type
Sirius	Canis Major	-1.47	1.42	8.6	Blue-white main-sequence
Canopus	Carina	-0.72	-5.53	310	White giant
Rigel Kentaurus	Centaurus	-0.28	4.07	4.4	Yellow main-sequence
Arcturus	Boötes	-0.10	-0.31	37	Orange giant
Vega	Lyra	0.03 (variable)	0.58	25	Blue-white main-sequence
Capella	Auriga	0.08	-0.48	43	Yellow giant
Rigel	Orion	0.13 (variable)	-7.92	860	Blue supergiant
Procyon	Canis Minor	0.40	2.68	11	White main-sequence
Achernar	Eridanus	0.50	-2.77	144	Blue-white main-sequence
Betelgeuse	Orion	0.45 (variable)	-5.14	498	Red supergiant
Hadar	Centaurus	0.61 (variable)	-5.23	390	Blue-white giant
Altair	Aquila	0.76 (variable)	2.20	17	White main-sequence
Acrux	Crux	0.77	-4.19	322	Blue-white subgiant
Aldebaran	Taurus	0.87	-0.63	67	Red giant
Spica	Virgo	0.98 (variable)	-3.55	250	Blue-white giant
Antares	Scorpius	0.90 (variable)	-5.28	550	Orange giant
Pollux	Gemini	1.16	1.09	34	Orange giant
Fomalhaut	Piscis Austrinus	1.16	1.73	25.0	Blue-white main-sequence
Mimosa	Crux	1.25 (variable)	-3.92	278	Blue-white giant
Deneb	Cygnus	1.25	-8.38	1,400	Blue-white supergiant